

WS #8 - Factor variables

Wednesday, September 30, 2026

DS 002R - Jo Hardin

Name: _____

Names of people you worked with: _____

Do you go to the mentor sessions? How can you get the most out of the mentor sessions?

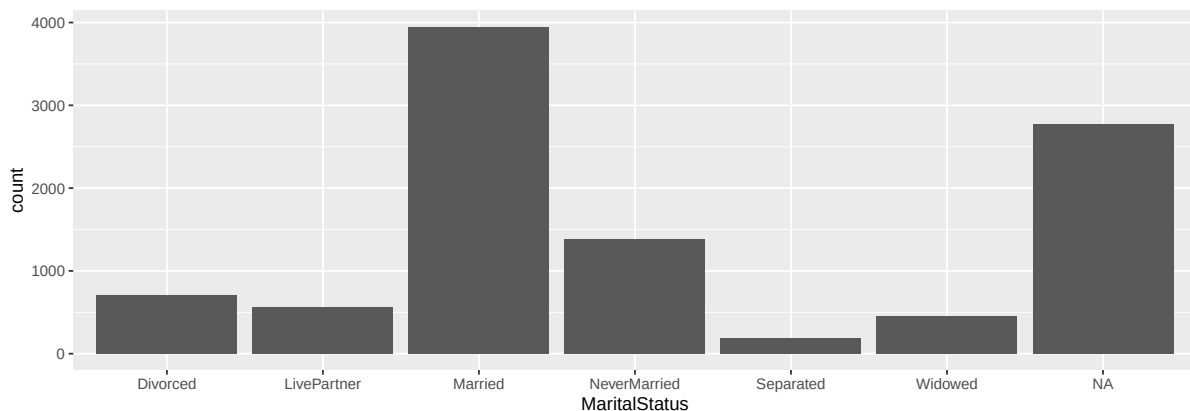
Task:

1. How would you order `MaritalStatus` in a more meaningful way? (No R code, just order the levels in a way that is meaningful to you.)
2. Fill in the last 5 columns of the table below (on back) as if for four new additional rows with `MaritalStatus` that contain the values `Divorced`, `LivePartner`, `Separated`, and `Widowed`?

```
select(NHANES, MaritalStatus) |> pull() |> levels()
```

```
[1] "Divorced"      "LivePartner"  "Married"      "NeverMarried"  
[5] "Separated"    "Widowed"
```

```
ggplot(NHANES, aes(x = MaritalStatus)) +  
  geom_bar()
```



```
NHANES |>
  select(MaritalStatus) |>
  sample_n(10) |>
  mutate(
    Marnum1 = as.numeric(MaritalStatus),
    Marnum2 = as.numeric(MaritalStatus) + 1,
    Marchar = as.character(MaritalStatus),
    Marchange1 = ifelse(MaritalStatus == "Married", "Lol", MaritalStatus),
    Marchange2 = if_else(MaritalStatus == "Married", "Lol", MaritalStatus),
  )
```

A tibble: 10 x 6

	MaritalStatus	Marnum1	Marnum2	Marchar	Marchange1	Marchange2
	<fct>	<dbl>	<dbl>	<chr>	<chr>	<chr>
1	NeverMarried	4	5	NeverMarr~	4	NeverMarr~
2	Married	3	4	Married	Lol	Lol
3	Married	3	4	Married	Lol	Lol
4	NeverMarried	4	5	NeverMarr~	4	NeverMarr~
5	<NA>	NA	NA	<NA>	<NA>	<NA>
6	Married	3	4	Married	Lol	Lol
7	Married	3	4	Married	Lol	Lol
8	<NA>	NA	NA	<NA>	<NA>	<NA>
9	Married	3	4	Married	Lol	Lol
10	Married	3	4	Married	Lol	Lol

Solution:

1. I'm not sure that there is a single right answer, but one potential ordering is:

NeverMarried, LivePartner, Married, Separated, Divorced, Widowed

2. Here are two tables with the relevant information on the 4 new levels: Divorced, LivePartner, Separated, and Widowed.

```
set.seed(5)
NHANES |>
  select(MaritalStatus) |>
  sample_n(3) |>
  mutate(
    Marnum1 = as.numeric(MaritalStatus),
    Marnum2 = as.numeric(MaritalStatus) + 1,
    Marchar = as.character(MaritalStatus),
    Marchange1 = ifelse(MaritalStatus == "Married", "Lol", MaritalStatus),
    Marchange2 = if_else(MaritalStatus == "Married", "Lol", MaritalStatus)
  )
```

A tibble: 3 x 6

	MaritalStatus	Marnum1	Marnum2	Marchar	Marchange1	Marchange2
	<fct>	<dbl>	<dbl>	<chr>	<int>	<chr>
1	<NA>	NA	NA	<NA>	NA	<NA>
2	NeverMarried	4	5	NeverMarri~	4	NeverMarr~
3	LivePartner	2	3	LivePartner	2	LivePartn~

```
set.seed(14)
NHANES |>
  select(MaritalStatus) |>
  sample_n(10) |>
  mutate(
    Marnum1 = as.numeric(MaritalStatus),
    Marnum2 = as.numeric(MaritalStatus) + 1,
    Marchar = as.character(MaritalStatus),
    Marchange1 = ifelse(MaritalStatus == "Married", "Lol", MaritalStatus),
    Marchange2 = if_else(MaritalStatus == "Married", "Lol", MaritalStatus)
  )
```

A tibble: 10 x 6

	MaritalStatus	Marnum1	Marnum2	Marchar	Marchange1	Marchange2
--	---------------	---------	---------	---------	------------	------------

<fct>	<dbl>	<dbl>	<chr>	<chr>	<chr>
1 Married	3	4 Married	Lol	Lol	Lol
2 Married	3	4 Married	Lol	Lol	Lol
3 Married	3	4 Married	Lol	Lol	Lol
4 Separated	5	6 Separated	5	Separated	Separated
5 Divorced	1	2 Divorced	1	Divorced	Divorced
6 Married	3	4 Married	Lol	Lol	Lol
7 Married	3	4 Married	Lol	Lol	Lol
8 Married	3	4 Married	Lol	Lol	Lol
9 <NA>	NA	NA <NA>	<NA>	<NA>	<NA>
10 Widowed	6	7 Widowed	6	Widowed	Widowed